

Lab 5.1: Self-Hosting and Configuring Jenkins Using Docker

Objective

This lab introduces students to **setting up Jenkins as a self-hosted CI/CD server using Docker**. Students will install, run, and configure Jenkins **locally**, preparing it for use in subsequent MLOps labs.

Task 1: Create the Custom Jenkins Docker Image

This task prepares a **custom Jenkins image** that already contains all required tools for future MLOps pipelines.

Step 1.1: Create a Dockerfile.jenkins file

The custom Jenkins image is based on:

- jenkins/jenkins:lts

It **pre-installs**:

- Python 3
- pip and virtualenv
- jq (for JSON parsing)
- bc (for numeric comparison)
- Docker CLI

Step 1.2: Build the Custom Jenkins Image

Students must build the Jenkins image using the provided Dockerfile.

```
docker build -t <ROLL_NO>-jenkins -f Dockerfile.jenkins .
```

Replace <ROLL_NO> with your roll no.

Task 2: Run Jenkins Using the Custom Image

Students must run Jenkins using the custom image created in Task 1.

Step 2.1: Run Jenkins Container

```
docker run -d -u root \  
-p 8080:8080 -p 50000:50000 \  
-v /var/run/docker.sock:/var/run/docker.sock \  
-v jenkins_home:/var/jenkins_home \  
--name <ROLLNO>_jenkins \  
my-custom-jenkins
```

- Replace <rollno> with your actual roll number

Task 3: Verify Jenkins Container Status

```
docker ps
```

Task 4: Unlock Jenkins

Step 4.1: Access Jenkins UI

Open a browser and go to:

```
http://localhost:8080
```

You should see the **Unlock Jenkins** screen.

Step 4.2: Retrieve Initial Admin Password

```
docker exec <rollno>_jenkins cat  
/var/jenkins_home/secrets/initialAdminPassword
```

Task 5: Initial Jenkins Configuration

Students must:

1. Select **Install suggested plugins**
2. Wait for plugin installation to complete
3. Create the admin user using the **mandatory format**

Field	Value
Username	<student_roll_no>
Full Name	Student's full name
Email	Official college email ID
Password	Student-defined

Task 6: Familiarize and explore Jenkins Dashboard

Deliverables

Students must submit **screenshots** showing:

1. **Custom image build**
 - Output of docker images

2. **Running Jenkins container**
 - Output of docker ps
3. **Admin creation screen**
4. **Jenkins dashboard**

Screenshots must clearly show:

- Roll number
- Jenkins instance